

Credit Risk Modelling

Phase 1

This project focuses on building predictive models to assess and measure credit risk for loan applicants, leveraging advanced statistical techniques and machine learning algorithms.



Project Objective & Success Criteria

Primary Objective

Build a robust predictive model that empowers the Risk Unit to accurately measure and assess credit risk across xyz Finance's loan portfolio. The model will serve as a critical decision-making tool for credit approvals and risk management strategies.

Success Criteria

- 1

Model Performance

AUC and Gini Coefficient must exceed 85%, demonstrating strong discriminatory power between default and non-default cases.
- 2

Statistical Significance

Kolmogorov-Smirnov statistic greater than 40, indicating clear separation between risk distributions.
- 3

Early Detection

Maximum KS statistic must appear within the first 3 deciles, ensuring early identification of high-risk borrowers.
- 4

Business Value

Model must be interpretable and actionable for business stakeholders and risk analysts.

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Historical loan portfolio data from xyz Finance spanning 27 months of customer and loan performance information.

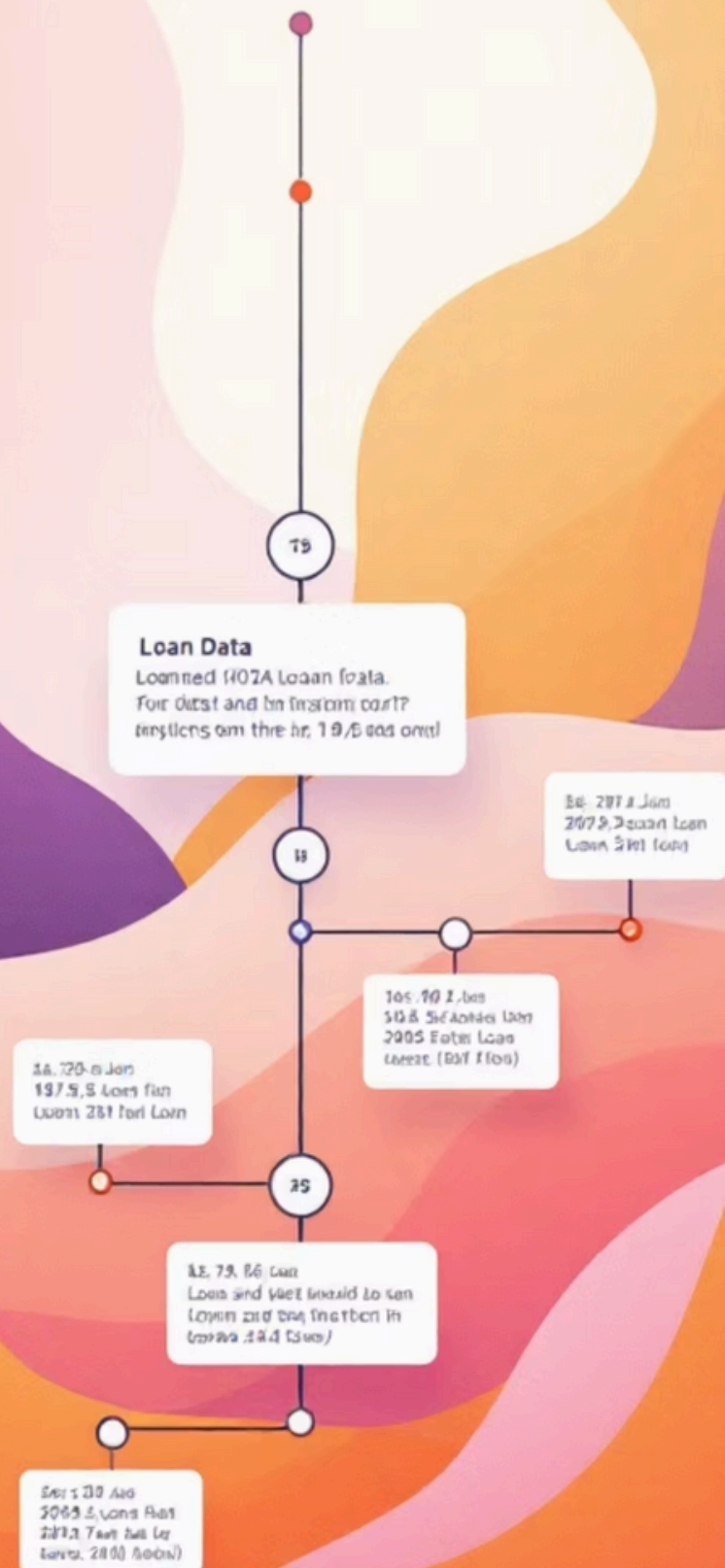
Historical loan portfolio data from xyz Finance spanning 27 months of customer and loan performance information.

February 2022 to February 2024

24 months of data used for model training and validation, capturing diverse economic conditions and borrower behaviors.

March 2024 to May 2024

3 months of holdout data for unbiased performance evaluation, simulating real-world model deployment scenarios.



Features – Customer Demographics

Customer demographic features provide essential context about borrower characteristics, financial stability, and residential patterns. These variables help establish baseline risk profiles and identify demographic segments with varying default propensities.

cust_id	Unique customer identifier for tracking and linking across datasets
age	Customer age in years, indicating life stage and financial maturity
gender	Gender classification for demographic analysis
marital_status	Current marital status affecting financial obligations and stability
employment_status	Employment type and status indicating income stability
income	Annual income level in currency units, primary repayment capacity indicator
number_of_dependents	Count of financial dependents affecting disposable income
residence_type	Housing situation (owned, rented, mortgaged) indicating financial commitment
years_at_current_address	Residential stability metric measured in years
city	Current city of residence for geographic risk assessment
state	State-level location for regional economic pattern analysis
zipcode	Postal code enabling granular geographic segmentation

Features - Loan Characteristics

Loan-specific features capture the financial structure, terms, and performance metrics of each credit facility. These variables are crucial for understanding borrower leverage, repayment capacity, and loan performance dynamics throughout the lifecycle.

loan_id	Unique loan identifier for tracking individual credit facilities
cust_id	Customer identifier linking loans to borrower profiles
loan_purpose	Intended use of funds (education, home, business, personal, etc.)
loan_type	Classification of loan product type and structure
sanction_amount	Approved credit limit or maximum loan amount
loan_amount	Actual amount borrowed by the customer
net_disbursement	Final amount credited to customer account after fees and deductions
loan_tenure_months	Total repayment period in months
principal_outstanding	Current outstanding principal balance (book size)
bank_balance_at_application	Customer's bank balance at time of loan application
disbursal_date	Date when loan funds were disbursed
installment_start_dt	Date of first EMI payment
default	Target variable: Binary indicator of loan default status (0 = No Default, 1 = Default)

Features - Credit Bureau Data

Credit bureau features provide comprehensive historical credit behavior data, offering critical insights into borrower payment discipline, credit appetite, and financial management patterns. These variables are among the strongest predictors of future default risk.

no_of_open_accounts

Total count of currently active credit accounts across all lenders, indicating current credit exposure

no_of_closed_accounts

Historical count of closed credit facilities, reflecting credit history depth and past borrowing patterns

total_loan_months

Cumulative duration of all credit facilities in months, measuring total credit experience

delinquent_months

Total months with payment delays across all accounts, a key indicator of payment discipline

total_dpd

Total days past due accumulated across all accounts and time periods

enquiry_count

Number of credit inquiries made by lenders, indicating credit-seeking behavior

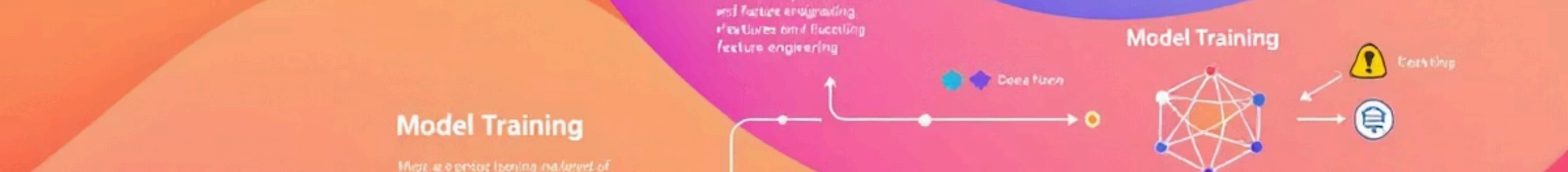
credit_utilization_ratio

Percentage of available credit currently in use, measuring credit dependency and financial stress

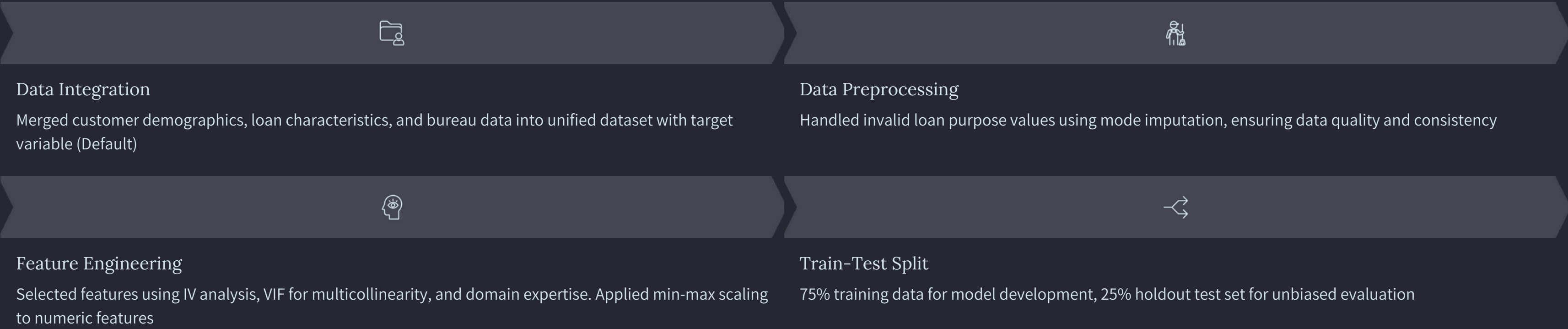
Significant Variables Driving Default Risk

Information Value (IV) analysis reveals the predictive power of each variable. Variables with IV > 0.3 are considered strong predictors, while IV between 0.1-0.3 indicates medium predictive strength. Our analysis identified several critical risk drivers.

Variable	IV Score	Risk Inference
credit_utilization_ratio	2.35	Higher usage of available credit significantly increases default risk, indicating financial stress
delinquency_ratio	0.71	Higher delinquency rates are strongly linked to increased default probability and poor payment discipline
loan_to_income	0.47	Higher loan amounts relative to income increase likelihood of default due to repayment burden
avg_dpd_per_delinquency	0.40	Higher days past due per delinquency correlates with severe payment difficulties
loan_purpose	0.36	Certain loan purposes (consumption vs. investment) show distinct default patterns
residence_type	0.24	Residence type has moderate impact, with renters showing higher risk than homeowners
loan_tenure_months	0.21	Longer loan tenures increase default risk due to extended exposure period
loan_type	0.16	Different loan types have minor but measurable influence on default patterns
age	0.08	Age shows minimal effect, with slight elevation in risk at extreme age ranges
number_of_open_accounts	0.08	More open accounts can indicate over-leverage and higher default risk



Modeling Methodology & Pipeline



Model Training Approach

- **Logistic Regression:** Baseline interpretable model
- **XGBoost:** Gradient boosting for complex patterns
- **Random Forest:** Ensemble method for robustness

Hyperparameter Optimization

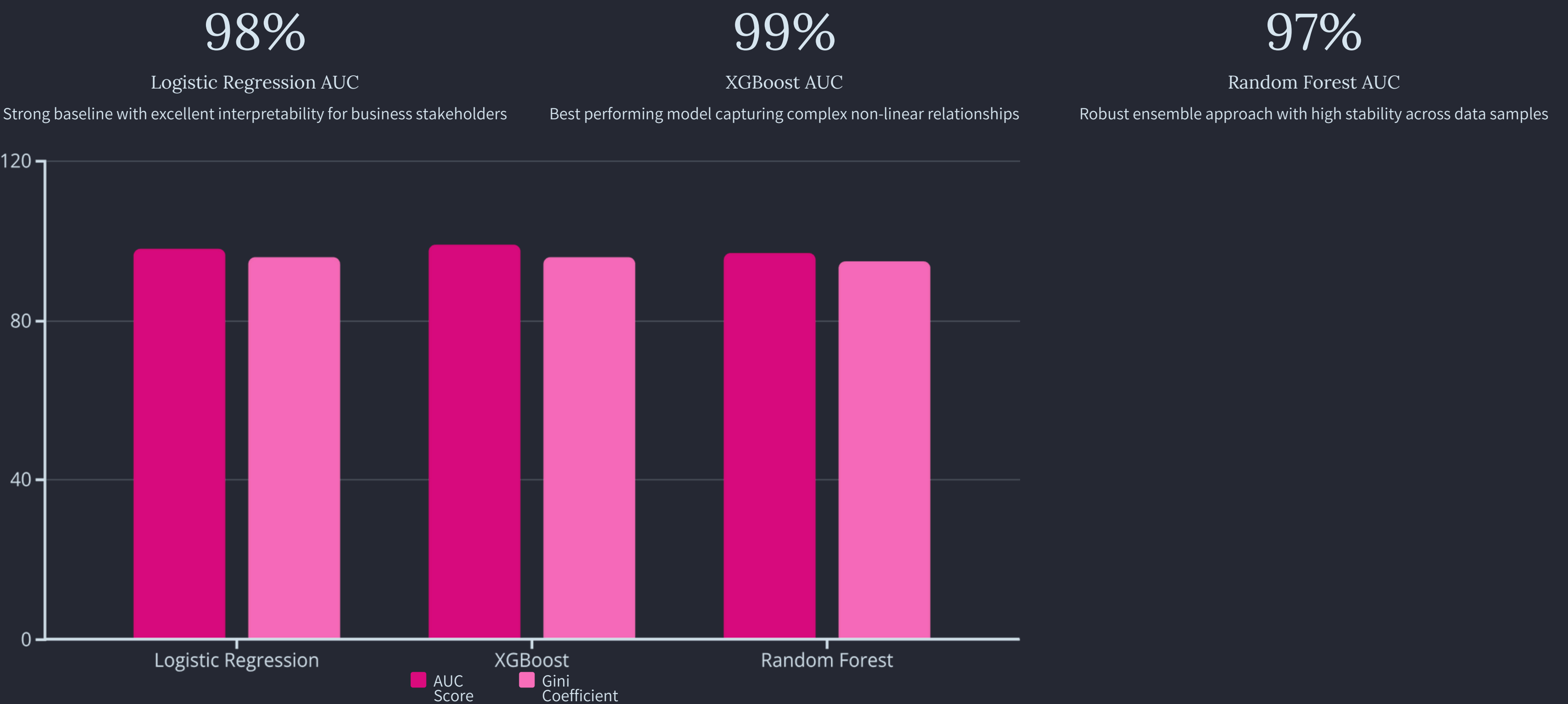
- **RandomizedSearchCV:** Efficient parameter space exploration
- **Optuna:** Advanced Bayesian optimization framework

Evaluation Metrics

- AUC-ROC, KS Statistic, Gini Coefficient
- Classification report with precision and recall

Model Performance Comparison

All three models demonstrated exceptional performance, significantly exceeding our success criteria. The consistency across models validates the quality of our feature engineering and data preprocessing pipeline.

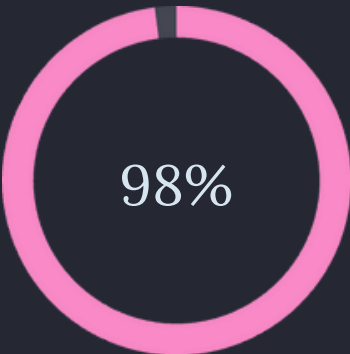


All models exceeded the 85% threshold for both AUC and Gini, with XGBoost achieving near-perfect discrimination. The Gini coefficients above 95% indicate exceptional model concentration and risk ranking capability.

Final Model Evaluation Results

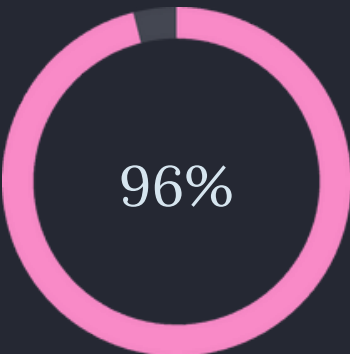
	Decile	Minimum Probability	Maximum Probability	Events	Non-events	Event Rate	Non-event Rate	Cum Events	Cum Non-events	Cum Event Rate	Cum Non-event Rate	KS
0	9	0.812	1.000	900.000	350.000	72.000	28.000	900.000	350.000	83.799	3.064	80.735
1	8	0.198	0.812	159.000	1091.000	12.720	87.280	1059.000	1441.000	98.603	12.615	
2	7	0.026	0.198	10.000	1239.000	0.801	99.199	1069.000	2680.000	99.534	23.461	76.073
3	6	0.004	0.026	5.000	1245.000	0.400	99.600	1074.000	3925.000	100.000	34.361	65.639
4	5	0.001	0.004	0.000	1249.000	0.000	100.000	1074.000	5174.000	100.000	45.295	54.705
5	4	0.000	0.001	0.000	1250.000	0.000	100.000	1074.000	6424.000	100.000	56.237	43.763
6	3	0.000	0.000	0.000	1250.000	0.000	100.000	1074.000	7674.000	100.000	67.180	32.820
7	2	0.000	0.000	0.000	1249.000	0.000	100.000	1074.000	8923.000	100.000	78.114	21.886
8	1	0.000	0.000	0.000	1250.000	0.000	100.000	1074.000	10173.000	100.000	89.057	10.943
9	0	0.000	0.000	0.000	1250.000	0.000	100.000	1074.000	11423.000	100.000	100.000	0.000

Outstanding Performance



AUC Score

Exceptional discrimination power



Gini Coefficient

Superior risk concentration



Top 3 Decile Capture

Outstanding early detection

✓ All Success Criteria Exceeded

The model significantly surpassed all established success criteria: AUC (98%) and Gini (96%) well above the 85% threshold, KS statistic exceeding 40, and remarkable 99.53% capture rate in top 3 deciles demonstrating exceptional early risk identification capability. The model is production-ready and will provide substantial value to Lauki Finance's risk management framework.